

§5 Substitution Rule

* The Substitution Rule

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x)) g'(x) dx = F(g(x)) + C.$$

where $F' = f$.

(pf) Let $u = g(x)$.

Then the differential du is

$$du = g'(x) dx.$$

$$\therefore \int f(g(x)) g'(x) dx = \int f(u) du$$

$$= F(u) + C$$

$$= F(g(x)) + C$$

$$\text{(cf)} \quad \int f(g(x)) g'(x) dx = \int f(g(x)) d(g(x))$$

$$= F(g(x)) + C$$

Exam 1 Find $\int x^3 \cos(x^4 + 2) dx$.

$$\begin{aligned} \text{(sol)} \quad \int x^3 \cos(x^4 + 2) dx &= \frac{1}{4} \int \cos(x^4 + 2) d(x^4 + 2) \\ &= \frac{1}{4} \sin(x^4 + 2) + C \end{aligned}$$

Exam 2 Evaluate $\int \sqrt{2x+1} dx$.

$$\begin{aligned} \text{(sol)} \quad \int \sqrt{2x+1} dx &= \frac{1}{2} \int (2x+1)^{1/2} d(2x+1) \\ &= \frac{1}{2} \frac{2}{3} (2x+1)^{\frac{3}{2}} + C \\ &= \frac{1}{3} \sqrt{(2x+1)^3} + C \end{aligned}$$

Exam 3 Find $\int \frac{x}{\sqrt{1-4x^2}} dx$

$$\begin{aligned} \text{(sol)} \quad \int \frac{x}{\sqrt{1-4x^2}} dx &= -\frac{1}{8} \int (1-4x^2)^{-\frac{1}{2}} d(1-4x^2) \\ &= -\frac{1}{8} 2(1-4x^2)^{\frac{1}{2}} + C \\ &= -\frac{1}{4} (1-4x^2)^{\frac{1}{2}} + C \end{aligned}$$

$$\begin{aligned} * \quad \int f(ax+b) dx &= \frac{1}{a} \int f(ax+b) d(ax+b) \\ &= \frac{1}{a} F(ax+b) + C \end{aligned}$$

Exam 4 Calculate $\int e^{5x} dx$.

$$\begin{aligned} \text{(sol)} \quad \int e^{5x} dx &= \int \frac{1}{5} e^{5x} d(5x) \\ &= \frac{1}{5} e^{5x} + C \end{aligned}$$

$$\begin{aligned}
 * \int \frac{f'(x)}{f(x)} dx &= \int \frac{1}{f(x)} d(f(x)) \\
 &= \ln |f(x)| + C
 \end{aligned}$$

*** Indefinite Integral of Trigonometric Functions**

$$1. \int \sin x \, dx = -\cos x + C$$

$$2. \int \cos x \, dx = \sin x + C$$

$$\begin{aligned}
 3. \int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx \\
 &= - \int \frac{-\sin x}{\cos x} \, dx \\
 &= -\ln |\cos x| + C \\
 &= \ln |\sec x| + C
 \end{aligned}$$

$$\begin{aligned}
 4. \int \cot x \, dx &= \int \frac{\cos x}{\sin x} \, dx \\
 &= \ln |\sin x| + C
 \end{aligned}$$

$$\begin{aligned}
 5. \int \sec x \, dx &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx \\
 &= \ln |\sec x + \tan x| + C
 \end{aligned}$$

$$\begin{aligned}
 6. \int \csc x \, dx &= \int \frac{\csc^2 x - \csc x \cot x}{\csc x - \cot x} \, dx \\
 &= \ln |\csc x - \cot x| + C
 \end{aligned}$$

*** Indefinite Integral of Hyperbolic Functions**

$$1. \int \sinh x \, dx = \cosh x + C$$

$$2. \int \cosh x \, dx = \sinh x + C$$

$$\begin{aligned} 3. \int \tanh x \, dx &= \int \frac{\sinh x}{\cosh x} \, dx \\ &= \ln(\cosh x) + C \end{aligned}$$

$$\begin{aligned} 4. \int \coth x \, dx &= \int \frac{\cosh x}{\sinh x} \, dx \\ &= \ln|\sinh x| + C \end{aligned}$$

$$\begin{aligned} 5. \int \operatorname{sech} x \, dx &= \int \frac{1}{\cosh x} \, dx \\ &= \int \frac{\cosh x}{\cosh^2 x} \, dx \\ &= \int \frac{1}{1 + \sinh^2 x} \, d(\sinh x) \\ &= \tan^{-1}(\sinh x) + C \end{aligned}$$

$$\begin{aligned} 6. \int \operatorname{csch} x \, dx &= \int \frac{\operatorname{csch}^2 x - \operatorname{csch} x \coth x}{\operatorname{csch} x - \coth x} \, dx \\ &= \ln|\operatorname{csch} x - \coth x| + C \end{aligned}$$

*** The Substitution Rule for Definite Integrals**

$g' : \text{continuous on } [a, b]$.

$f : \text{continuous on the range of } u = g(x)$.

$$\Rightarrow \int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

(pf) Let F be an antiderivative of f .

$$\begin{aligned} \Rightarrow \int_a^b f(g(x)) g'(x) dx &= F(g(x)) \Big|_a^b \\ &= F(g(b)) - F(g(a)) \\ &= F(u) \Big|_{g(a)}^{g(b)} \\ &= \int_{g(a)}^{g(b)} f(u) du \end{aligned}$$

Exam 6 Evaluate $\int_1^2 \frac{dx}{(3-5x)^2}$.

$$\begin{aligned} \text{(sol)} \quad \int_1^2 \frac{dx}{(3-5x)^2} &= \int_1^2 \frac{dx}{(5x-3)^2} \\ &= \int_1^2 (5x-3)^{-2} dx \\ &= \left. \frac{1}{5}(-1)(5x-3)^{-1} \right|_1^2 \\ &= -\frac{1}{5} \left(\frac{1}{7} - \frac{1}{2} \right) \\ &= \frac{1}{14} \end{aligned}$$

Exam 7 Calculate $\int_1^e \frac{\ln x}{x} dx$.

$$\begin{aligned} \text{(sol)} \quad \int_1^e \frac{\ln x}{x} dx &= \int_1^e \ln x \, d(\ln x) \\ &= \left. \frac{1}{2}(\ln x)^2 \right|_1^e \\ &= \frac{1}{2} \end{aligned}$$

* Integrals of Symmetric Functions

f : continuous on $[-a, a]$.

$$\Rightarrow \text{(a) } f: \text{ even } [f(-x) = f(x)] \Rightarrow \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx .$$

$$\text{(b) } f: \text{ odd } [f(-x) = -f(x)] \Rightarrow \int_{-a}^a f(x) dx = 0 .$$

(pf) Let $x = -u \Rightarrow dx = -du$ and $x = -a \rightarrow u = a, \quad x = 0 \rightarrow u = 0$

$$\begin{aligned} \therefore \int_{-a}^0 f(x) dx &= \int_a^0 f(-u) (-du) \\ &= \int_0^a f(-u) du \\ &= \int_0^a f(-x) dx \end{aligned}$$

$$\begin{aligned} \Rightarrow \int_{-a}^a f(x) dx &= \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \\ &= \int_0^a f(-x) dx + \int_0^a f(x) dx \\ &= \int_0^a [f(-x) + f(x)] dx \end{aligned}$$

(a) Suppose that f is even, that is, $f(-x) = f(x)$.

$$\begin{aligned} \Rightarrow \int_{-a}^a f(x) dx &= \int_0^a [f(x) + f(x)] dx \\ &= 2 \int_0^a f(x) dx \end{aligned}$$

(b) Suppose that f is odd, that is, $f(-x) = -f(x)$.

$$\Rightarrow \int_{-a}^a f(x) dx = \int_0^a [-f(x) + f(x)] dx = 0$$

Exam 8 Let $f(x) = x^6 + 1$.

Then $f(x) = x^6 + 1$ is even.

$$\begin{aligned}\Rightarrow \int_{-2}^2 f(x) dx &= 2 \int_0^2 (x^6 + 1) dx \\ &= \frac{284}{7}\end{aligned}$$

Exam 9 Let $f(x) = \frac{\tan x}{1 + x^2 + x^4}$.

Then $f(x) = \frac{\tan x}{1 + x^2 + x^4}$ is odd.

$$\Rightarrow \int_{-1}^1 f(x) dx = \int_{-1}^1 \frac{\tan x}{1 + x^2 + x^4} dx = 0.$$